INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – I , First Semester Examination, August- 2025

Subject Code: BCACC103

Subject Name: Discrete Structure

Full Marks: 30 Duration: 1 Hour

Part - A

Answer any 5 questions out of 6

Each question carries 2 marks (2 × 5)

1. If A={1,2,3,4,5,6,7,8,9}, B={2,3,4,5}, C={2,4,6,8} and D={4,5,6,7}, find $B \cup C$ and $B \cap D$. 2

2. If $U = \{x : x \in Z \text{ and } 1 \le x \le 10\}$ and $A = \{x : x \in U \text{ and } x \text{ is a prime number}\}$, $B = \{x : x \in U \text{ and } x \text{ is even}\}$, then find $A \Delta B$. 2

3. If A={1,2,3}, B={2,3,4}, write down the following set $\{(x, y) : (x, y) \in A \times B \text{ and } x < y\}$ 2

4. Define regular graph with proper example. 2

5. Draw a graph which is not simple, having two isolated vertices. 2

6. If A={1,2}, B={2,3}, C={3,4}, then find $(A \times B) \cap (A \times C)$. 2

Part - B

Answer any 2 questions out of 3

Each question carries 5 marks (5 × 2)

7. Let A,B,C be subsets of a Universal set U. Prove that 5

$$A \times (B \cup C) = (A \times B) \cup (A \times C)$$

8. Write a short note on Spanning Sub- graph. 5

9. Write a short note on Hamiltonian Graph. 5

Part - C

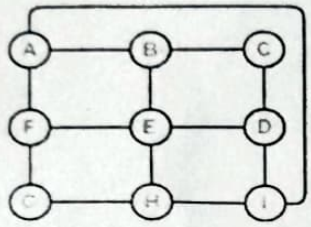
Answer any 1 question out of 2

Each question carries 10 marks (10 × 1)

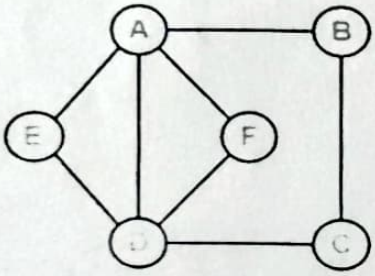
10. In a survey of 320 persons, number of persons taking tea is 210, taking milk 100 and coffee is 70. Number of persons who take tea and milk is 50, milk and coffee is 30, tea and coffee is 50. The number of persons taking all the three together is 20. Summarize the value which gives the number of persons who take neither tea nor milk nor coffee. **10**

11. Check whether the following graphs are Euler or not, give proper reason for your answer : **10**

i



ii.



--End--